LEARN CODING

ale66

github.com/ale66/learn-coding

RESOURCES

- Online materials
- install components ⬇️

github.com/ale66/learn-coding

ONLINE MATERIALS

- Moodle (MyAriel) web page for static information
- MS Teams channel
- online code execution with visualisation
- GitHub presence

MS TEAMS CHANNEL

Live broadcast and recording from the MS Teams team.

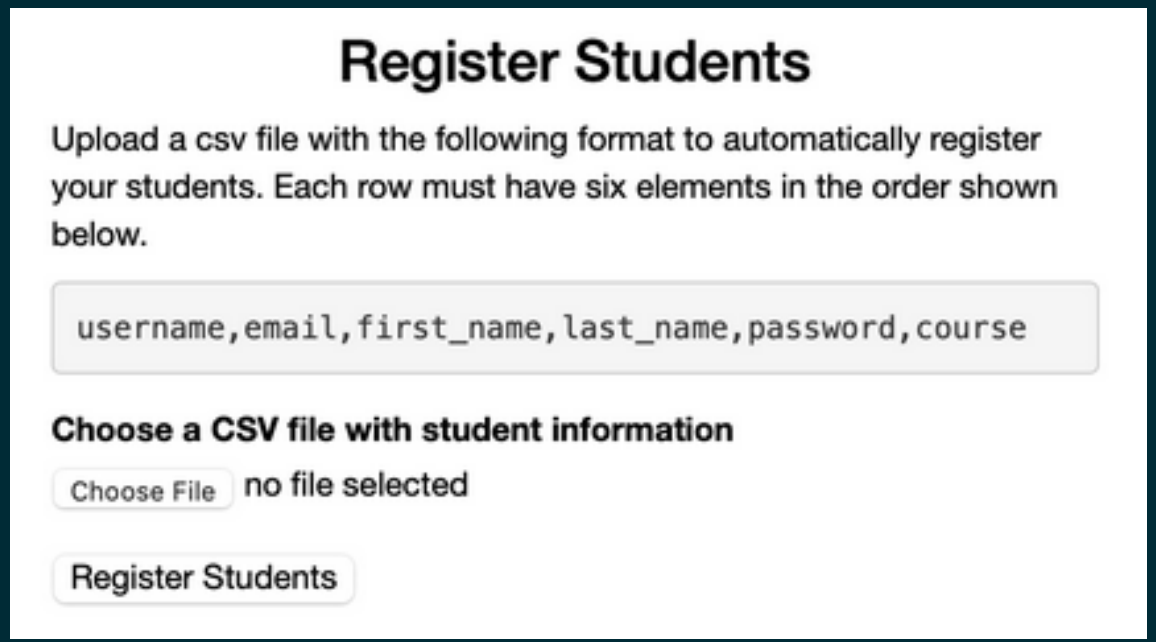
DM on Teams to request inclusion

ONLINE TEXTBOOK

For reference: *Foundations of Python Programming*, a version of **Think Python**, from

Runestone academy

To join, please write me on Teams with these informations



github.com/ale66/learn-coding

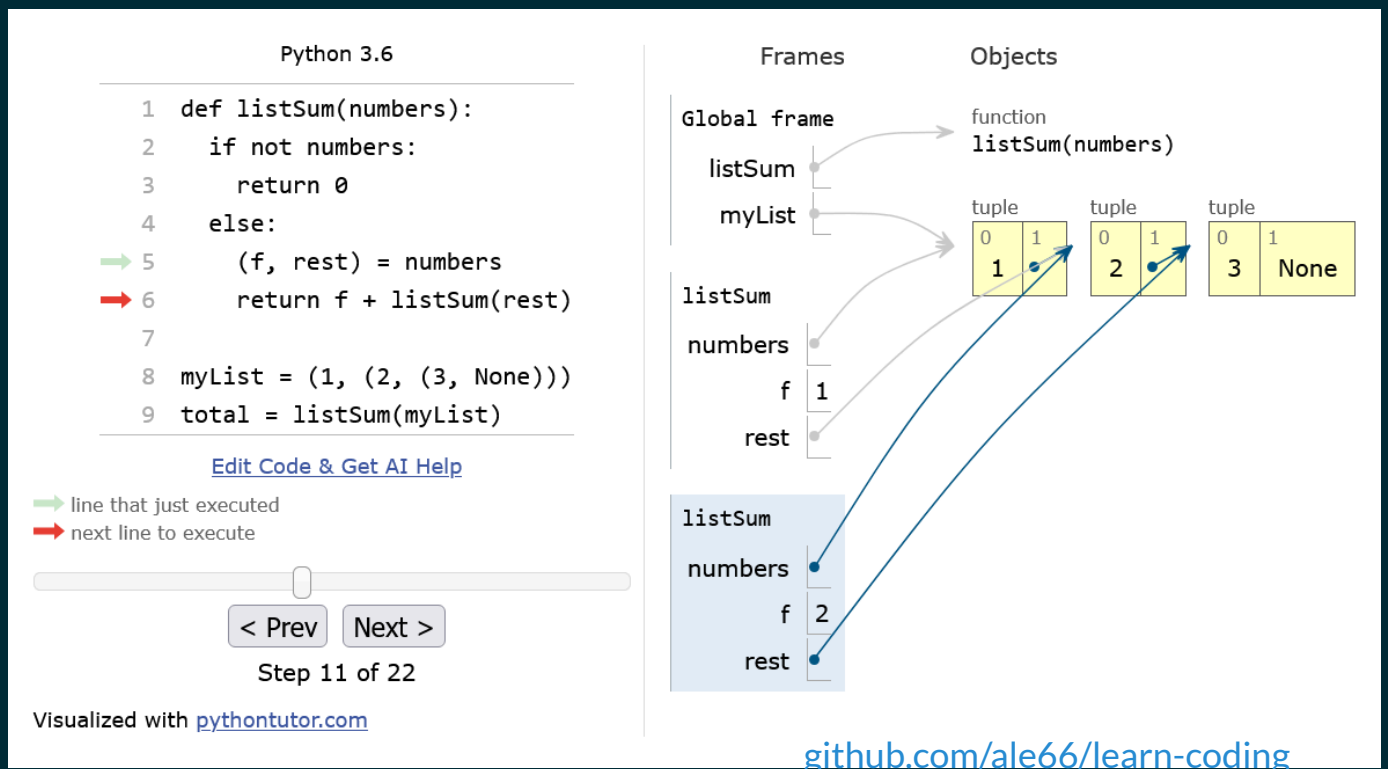
GITHUB PRESENCE

A public repository with most materials is on github.com/ale66/learn-coding

PYTHONTUTOR

Step-by-step execution of simple codes with a nice visualisation of what happens in the computer memory: a crucial tool for learning

pythontutor.com



github.com/ale66/learn-coding

INSTALL COMPONENTS

INSTALL NEEDED COMPONENTS

For Win 11: you may need to authorise installations.

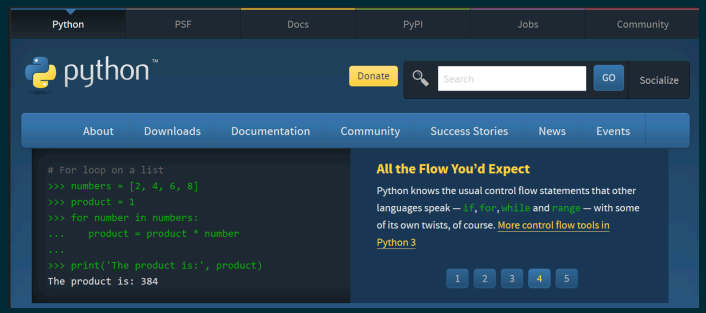
For MacOS: proceed to normal app download+installation.
In the notes below substitute

Windows Power Shell by *terminal*

dir by *ls -lrt*

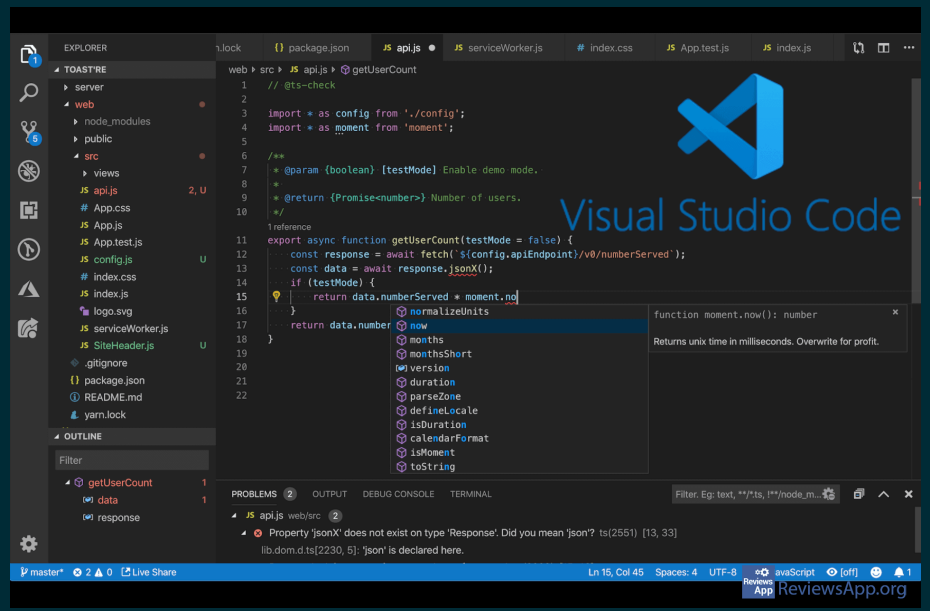
For Linux: you don't need me 😄

PYTHON



- install from www.python.org/
- no need for Anaconda virtual env. (yet)

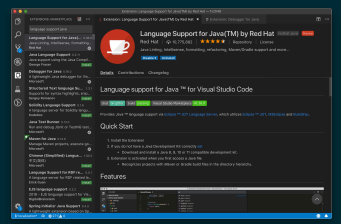
VS CODE



- Free and light
- lots of extensions allow customisation

github.com/ale66/learn-coding

VS CODE EXTENSIONS



CTRL+Shift+x to the markeplace

- Python, by Microsoft
- Jupyter, by Microsoft
- Markdown All-in-one
- Simple CSV editor

PIP

an external command for your Terminal/Powershell

fetch+install *packages* containing needed Py. functions

Now it's part of the distribution, read below for reference

Get **get-pip.py** from the PIP web site.

```
1  > wget https://bootstrap.pypa.io/get-pip.py -o get-pip.py
```

```
1  > python get-pip.py
```

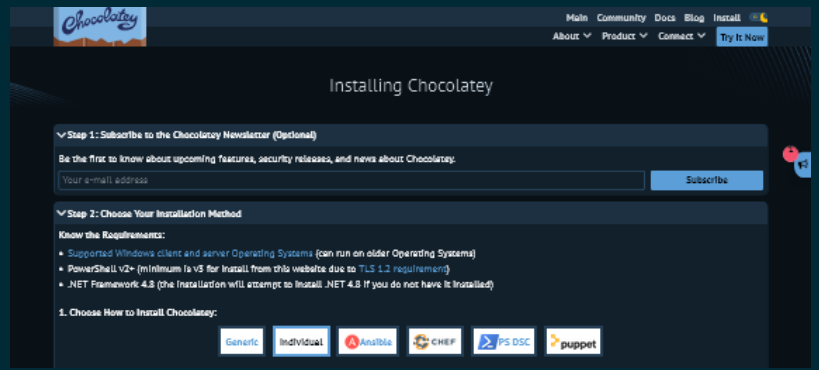
GET ESSENTIAL MODULES

```
> pip install os
...
> pip install numpy
...
> pip install pandas
...
> pip install matplotlib
```

APP MANAGEMENT

- for Win: Chocolatey
- for iOs: Homebrew

CHOCOLATEY



It manages installed software as linux does

Intallation is of average complexity

Start a **PowerShell** in administrator mode;

see ‘individual installation’.

EXAMPLE COMMANDS

```
> choco install python
```

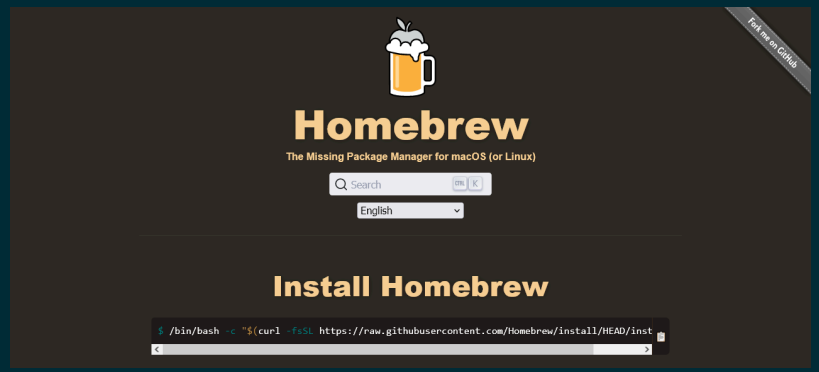
```
> choco install vscode
```

```
> choco install pip
```

```
> choco upgrade python
```

```
# upgrade all installed packages
> choco upgrade all --yes
```

HOME BREW



Essentially the same as Chocolatey, but for MacOS.

```
1  > brew --cask install visual-studio-code
```

N.B. to install Git on MacOs we need brew



- a protocol for managing code (web pages *are* code)
- keeps track of our work so it can undo changes that disrupt previous work
- collaboration is supported so teams can work asynchronously
- branching allows to explore new ideas without spoiling the existing code



GITHUB

Github is a community repository for sharing code and open collaborations

download the Git app

to maintain your copy of the GH repo github.com/ale66/learn-web up-to-date

General tendence: code is public but data is very private

INCREMENTAL/DIFFERENTIAL SYNCHRONISATION

it's easy to synchronise your local copy of a repo with the Github version

- first time: clone existing repository into a suitable folder

```
1  >git clone https://github.com/ale66/learn-coding.git
```

- updating: from a terminal working on the repository folder (or below)

```
1  >git pull
```

FURTHER TIPS

- try practicing daily, in your **zone**
- try getting past frustation and *enjoy* the progress you make: you are in control now
- sign up to Stack Overflow to find solutions
- *do not*, at any rate, watch Python videos on YouTube…